COURSE SYLLABUS for SIP10

I. VISION	Inspired by the teachings and tradition of the Catholic Church, The Angeles University Foundation envisions to emerge as a Center of excellence in instruction, research, and community extension services in the region and in the global community.
II. MISSION	To realize its vision, the university is committed to the "total development of man for God and humanity.
III. CORE VALUES	**Mabuti** (Inner Goodness). Initiate activities that will inspire the university-community to become authentic witnesses of the Christian Faith. **Magaling** (Excellence). Provide venues and opportunities for the university-community to become critical and analytical in issues that hinder the promotion of human dignity. **May Malasakit sa Kapwa** (Genuine Concern for Others). Prepare men and women who can translate their Christian principles into genuine compassion for others especially to the least, the last and the lost.
IV. INSTITUTIONAL LEARNING OUTCOMES	A. Values-Oriented (VO) B. Socially and Ethically Responsible (SER) C. Professionally Competent (PC) D. Critical and Creative Thinker (CCT) E. Lifelong Learner (LL) F. Globally-Oriented (GO)
V. COURSE DETAILS	**Course Code:** SIP10 **Course Title:** Social Issues and Professional Ethics **Course Type:** Lecture **Credit Units:** 3 units **Contact Hours:** 3 hours per week **Prerequisites:** None **Mode of Delivery:** In Person **Course Description:** This course covers essential topics in Social Issues and Professional Ethics. Topics include Definition of Ethics, Data Science Ethics, History and Concept of Informed Consent, Data Ownership, Privacy, Anonymity, Data Validity, Algorithmic Fairness, Societal Consequences and Code of Ethics. It provides case studies with concrete examples of situations involving moral or ethical questions. It explains variety of tools to help make ethically and morally complex decisions in situations encountered throughout their professional careers.

VI. DESCRIPTION OF THE TERMINAL REQUIREMENT

MCO1: A research-based written paper where students critically examine a contemporary issue in computing ethics (e.g., AI ethics, data privacy, cybersecurity, social media ethics).
MCO2: Conduct a seminar related to Social Issues and Professional Ethics.

VII. PROGRAM LEARNING OUTCOMES (PLOs)

B.S. Computer Science

1. Communicate effectively with the computing community with regards to current trends;
2. Make effective presentations to a range of audiences about technical problems and their solutions;
3. Recognize the legal and ethical standards in the utilization of computing technology;
4. Evaluate algorithms, to select from a range of possible options, to provide justification for that selection, and to implement the algorithm in a particular context;
5. Apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of computer-based systems;
6. Evaluate, design and develop a solution through software development and/or exploratory/experimental research for a computing problem;
7. Identify all of the data, information, and knowledge elements and related organizations, for a computational science application;
8. Work effectively and independently in multidisciplinary and multi-cultural teams.

B.S. Information Technology

1. Analyze the local and global impact of technology on individuals, organization, and society;
2. Utilize current IT techniques and technologies;
3. Design and manage database systems;
4. Monitor and maintain computer systems and networks;
5. Design and implement secured IT infrastructure;
6. Develop, implement and maintain computer systems based on user requirements;
7. Communicate effectively using technical term;
8. Work effectively in teams to accomplish a common goal;
9. Adhere to standard of ethics including relevant industry and organizational codes of conduct; and
10. Engage in continuing professional education for self-improvement and updates.

VIII. COURSE LEARNING OUTCOMES (CLOs)

1. Demonstrate understanding of ethical theories and apply them in computing-related contexts.
2. Analyze the impact of computing on individuals, organizations, and society at large.
3. Identify and resolve ethical and legal issues related to computing using professional standards.
4. Apply knowledge of relevant social issues to decision-making in technology design and development.
5. Demonstrate professional and ethical responsibility in team-based projects and communication.

IX. CURRICULAR MAPPING

Fill the appropriate space with the color box to show alignment.

B. S. Computer Science

MAPPING OF PLO and ILO

PLO	VO	SER	PC	CCT	LL	GO
PLO1			■		■	■
PLO2			■		■	■
PLO3	■	■	■		■	■
PLO4			■	■		
PLO5			■	■		
PLO6			■	■		
PLO7			■	■		■
PLO8	■	■	■		■	■

MAPPING OF PLO and CLO

PLO	CLO1	CLO2	CLO3	CLO4	CLO5
PLO1	■	■			■
PLO2	■	■	■	■	■
PLO3	■	■	■	■	■
PLO4					
PLO5					
PLO6					
PLO7		■		■	■
PLO8		■	■	■	■

B. S. Information Technology

MAPPING OF PLO and ILO

PLO	VO	SER	PC	CCT	LL	GO
PLO1			■		■	■
PLO2			■			
PLO3	■		■	■		
PLO4			■		■	
PLO5	■	■				■
PLO6			■	■		
PLO7			■	■		
PLO8			■	■		
PLO9			■	■		
PLO10			■	■		

MAPPING OF PLO and CLO

PLO	CLO1	CLO2	CLO3	CLO4	CLO5
PLO1	■	■	■	■	■
PLO2					
PLO3					
PLO4					
PLO5					
PLO6					
PLO7	■	■	■	■	■
PLO8	■	■	■	■	■
PLO9	■	■	■	■	■
PLO10	■	■	■	■	■

X. TOPICS AND TEACHING-LEARNING ACTIVITIES

Topic	Objective	Hours/Week	CLO	Teaching-learning activities		Assessment
I. Introduction to Ethics and Professionalism A. Nature of Ethics B. Code of Ethics in Computing (ACM/IEEE) C. Definition and importance of ethics in computing D. Difference between morals, ethics, and laws	1. Definition and importance of ethics in computing 2. Difference between morals, ethics, and laws 3. ACM/IEEE Code of Ethics principles	6 hours 1st – 2nd week	1,2	**Teacher's activity**	• Explain ACM/IEEE code of ethics • Facilitate class discussion on ethics in computing • Provide case examples	**Recitation on ACM** Code of Ethics. Rubric on Ethics **poster infographics** (Ethics in Everyday Technology). **Role Play ACM** Code of ethics **Quiz** on Introduction to Ethics and professionalism Ethics
				Onsite students' activity	• Write a reflection on "Why Ethics Matters in IT" • Group sharing: Ethical dilemmas in daily tech use • Create posters summarizing ACM/IEEE principles	
II. Responsible Artificial Intelligence A. Principles of Responsible AI B. AI Bias and Fairness C. Explainable and Transparent AI D. AI Risk Management E. Human-Centered AI	1. Define Responsible AI concepts. 2. Identify key stakeholders in AI ecosystems. 3. Explain the importance of responsible AI practices.	6 hours 3rd – 4th week	1,3,4	**Teacher's activity**	• Explain the principles of Responsible AI using examples and short videos. • Discuss AI bias and fairness using real-world case studies. • Present examples of Responsible AI in	**Peer Evaluation** (Review classmates' Responsible AI proposals and provide constructive feedback.) **Group Presentation**

F. Responsible AI Across Industries G. Responsible AI Lifecycle				<table><tr><td></td><td>healthcare, business, education, and finance.</td></tr><tr><td>**Onsite students' activity**</td><td>• Participate in discussion and identify examples of Responsible AI in daily life. • Analyze a case study and identify possible AI biases. • Compare how different industries use AI responsibly.</td></tr></table>	Present how a specific industry (healthcare, education, finance, retail, etc.) applies Responsible AI practices. **Role-Playing Activity** Assume the roles of AI developers, business managers, customers, and regulators to discuss ethical AI decisions.
III. **Foundations of AI Ethics** A. Introduction to Ethics B. Ethical Theories and Frameworks a. Utilitarianism b. Deontology c. Virtue Ethics C. Ethical Decision-Making Models D. AI and Moral Responsibility E. Human Rights and AI	1. Apply ethical frameworks in AI scenarios. 2. Analyze ethical dilemmas in AI applications. 3. Develop ethical recommendations for responsible AI implementation based on recognized ethical principles and legal standards.	6 hours 5th – 6th week	2,3,4	<table><tr><td>**Teacher's activity**</td><td>• Present real-world AI ethical issues (e.g., biased AI, privacy concerns). • Present AI case studies and guide students in applying ethical theories. • Demonstrate the steps of an ethical decision-making model.</td></tr><tr><td>**Onsite students'**</td><td>• Complete a short</td></tr></table>	Pre-test to determine students' prior knowledge of ethics and AI. Active participation in discussions, debates, and classroom activities. Present recommendations for resolving an AI ethics issue based

				activity	reflection on the importance of ethics in technology. • Work in small groups to discuss ethical decisions. • Apply the ethical decision-making model to solve AI-related ethical issues.	on ethical frameworks and human rights principles.
IV. **Data Privacy, Security, and Legal Considerations** A. Data Privacy Principles B. Privacy by Design C. Data Protection Laws D. Philippine Data Privacy Act (RA 10173) E. AI Security Risks F. Cybersecurity and AI	1. Analyze privacy issues in AI applications. 2. Apply legal and regulatory requirements.	6 hours 7th – 8th week	3,4,5	**Teacher's activity**: • Invite a guest speaker (e.g., Data Protection Officer) • Present RA 10173 highlights • Analyze sample privacy policies **Onsite students' activity**: • Audit a mobile app's data permissions • Create a privacy guideline for students • Mock interview: Data privacy violations	After attending the talk from a Data Protection Officer, students write a reflection (300–500 words) Short Quiz Data Protection Prepare a 5–10 minute presentation summarizing the main sections of RA 10173, focusing on rights of data subjects, lawful processing, and penalties. Analyze a sample privacy policy from a website, app, or institution and evaluate it against

					RA 10173 compliance.
V. **Fairness, Bias, and Transparency in AI** A. Algorithmic Bias B. Sources of AI Bias C. Fairness Metrics D. Explainable AI (XAI) E. Transparency and Interpretability F. Responsible Data Collection	1. Explain the concepts of fairness, bias, and transparency in Artificial Intelligence. 2. Identify the different sources of algorithmic bias throughout the AI lifecycle. 3. Evaluate AI systems using appropriate fairness metrics. Analyze the importance of Explainable AI (XAI) in promoting trust, accountability, and responsible AI.	3 hours 9th week	3,4	**Teacher's activity** • Facilitate a class discussion on the ethical and societal impacts of biased AI. • Facilitate a brainstorming activity on reducing bias in AI systems. **Onsite students' activity** • Analyze case studies to identify instances of algorithmic bias. • Identify potential sources of bias in sample datasets and AI applications.	Bias Detection Exercise Active participation in discussions, case study analysis, group activities, and classroom exercises. Analyze a real-world AI application to identify bias, evaluate fairness, and recommend ethical improvements.
Midterm examinations					
VI. **AI Governance Frameworks** A. AI Governance Fundamentals B. Governance Structures C. Risk-Based AI Governance	1. Explain AI governance structures. 2. Develop organizational AI policies and auditing practices that support responsible and trustworthy AI. 3. Explain the concepts, principles, and importance of AI governance.	9 hours 10th – 12th week	2,3,4	**Teacher's activity** • Present real-world examples of AI governance in organizations. • Present organizational AI governance models.	Participate in class discussions on the importance of AI governance. Create a simple organizational chart illustrating AI governance roles.

<table>
<tr><td>D. AI Accountability Structures
E. Organizational AI Policies
F. AI Auditing, AI Assurance</td><td></td><td></td><td></td><td><table><tr><td></td><td>• Guide students through a risk assessment workshop.</td></tr><tr><td>Onsite students' activity</td><td>•</td></tr></table></td><td>Draft an organizational AI policy for a selected business or institution.</td></tr>
<tr><td>VII. Social Impact of Computing
A. Automation and job displacement
B. Bridging the digital divide (urban vs. rural access)
C. Tech sustainability and e-waste management</td><td>1. Identify societal issues affected by computing (e.g., automation, e-waste).
2. Assess the digital divide in the local or national context.
3. Recommend strategies for inclusive and sustainable tech use.</td><td>6 hours
13th – 14th week</td><td>1,3,4</td><td><table><tr><td>Teacher's activity</td><td>• Present dataset global and local case studies
• Facilitate impact mapping activity
• Share stories on automation and e-waste</td></tr><tr><td>Onsite students' activity</td><td>• Group activity: Tech impact infographic
• Write a position paper on AI replacing jobs
• Present solutions to digital divide problems</td></tr></table></td><td>Presentation Data analysis and Visualization.

Create an infographic showing positive and negative effects of technology.

Assess: Accuracy, visual clarity, and creativity.</td></tr>
<tr><td>IX. Legal Frameworks and Governance

A. Provisions and penalties under RA 8792 (E-Commerce</td><td>1. Analyze the penalties imposed for violations related to unlawful access, electronic data interference, and other prohibited acts under RA 8792.</td><td>6 hours
15th – 16th week</td><td>1,3,5</td><td><table><tr><td>Teacher's activity</td><td>• Invite a guest speaker (e.g., Company Policy Officer)</td></tr></table></td><td>Analyze a real cybercrime case under RA 8792 or RA 10175.
Assess: Law</td></tr>
</table>

Act) B. Cybercrime Prevention Act (RA 10175) basics C. Legal responsibilities of ISPs and tech companies	2. Apply the principles of responsible digital citizenship and cybersecurity awareness in compliance with cybercrime laws. 3. Analyze ethical and legal issues involving user privacy, data retention, and online platform accountability. 4. Evaluate the significance of the E-Commerce Act in promoting secure and reliable digital business transactions.			• Facilitate group think-pair-share • Discuss RA 8792, RA 10175 • Analyze tech company case studies **Onsite students' activity** • Panel discussion: "Should facial recognition be banned?" • Review a cybercrime case and present findings • Create a compliance checklist • Group report on e-commerce law violations	application, clarity, and ethical insight. Join a debate on "Should facial recognition be banned?" Assess: Argument strength, use of legal/ethical points, and teamwork. Create a checklist for businesses to comply with PH cyber laws. Assess: Completeness, relevance, and legal accuracy. Research and report on a case of e-commerce law violation. Assess: Case clarity, law applied, and team collaboration.
X. **Responsible AI Implementation** 1. AI Risk Assessment 2. AI Impact Assessment	1. Conduct AI risk assessments. 2. Develop AI implementation plans.	6 hours 17th – 18th week	1,4,5	**Teacher's activity** • Demonstrate how to identify, analyze, and prioritize AI risks	AI Risk Assessment Workshop AI Audit Checklist Development

3. Responsible AI by Design 4. Ethical AI Development Lifecycle 5. Governance Documentation Monitoring and Auditing AI Systems					using real-world examples. • Guide students in assessing the potential impacts of AI systems. • Present best practices and international AI governance standards.	Present a Responsible AI Implementation Plan for a selected AI application or organization. Performance Task Create a simple Responsible AI Implementation Plan that includes risk assessment, governance, and monitoring strategies.
				Onsite students' activity	• Perform a basic AI risk assessment using a risk matrix. • Analyze the potential impacts of an AI application on individuals, organizations, and society. • Create a flowchart illustrating the Ethical AI Development Lifecycle.	
Final examinations						

XI. GRADING SYSTEM

Midterm Period Raw=(Midterm Class Standing X 0.6) + (Midterm Exam X 0.4)
Midterm period raw is to be transmuted. The transmuted grade is encoded in School Automate as midterm grade.

Final Period Raw=(Final Class Standing X 0.5) + (Final Exam X 0.5)
Final Raw Grade=(Midterm Grade Raw +Final Period Raw)/2
Final raw grade is to be transmuted. The transmuted grade is encoded in School Automate as final grade.

XII. REFERENCES

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